Schrodinger equation would never have a macroscopic representation analogous to the macroscopic representation of the amplitude for photons. On the other hand, it is now realized that the phenomena of superconductivity presents us with just this situation.

21-5 Superconductivity

As you know, very many metals become superconducting below a certain temperature[6]—the temperature is different for different metals. When you reduce the temperature sufficiently the metals conduct electricity without any resistance. This phenomenon has been observed for a very large number of metals but not for all, and the theory of this phenomenon has caused a great deal of difficulty. It took a very long time to understand what was going on inside of superconductors, and I will only describe enough of it for our present purposes. It turns out that due to the interactions of the electrons with the vibrations of the atoms in the lattice, there is a small net effective *attraction* between the electrons. The result is that the electrons form together, if I may speak very qualitatively and crudely, bound pairs.

Now you know that a single electron is a Fermi particle. But a bound pair would act as a Bose particle, because if I exchange both electrons in a pair I change the sign of the wave function twice, and that means that I don't change anything. A pair is a Bose particle.

The energy of pairing-that is, the net attraction-is very, very weak. Only a tiny temperature is needed to throw the electrons apart by thermal agitation, and convert them back to "normal" electrons. But when you make the temperature sufficiently low that they have to do their very best to get into the absolutely lowest state; then they do collect in pairs.

I don't wish you to imagine that the pairs are really held together very closely like a point particle. As a matter of fact, one of the great difficulties of understanding this phenomena originally was that that is not the way things are. The two electrons which form the pair are really spread over a considerable distance; and the mean distance between pairs is relatively smaller than the size of a single pair. Several pairs are occupying the same space at the same time. Both the reason why electrons in a metal form pairs and an estimate of the energy given up in forming a pair have been a triumph of recent times. This fundamental point in the theory of superconductivity was first explained in the theory of Bardeen, Cooper, and Schrieffer,[7] but that is not the subject of this seminar. We will accept, however, the idea that the electrons do, in some manner or other, work in pairs, that we can think of these pairs as behaving more or less like particles, and that we can therefore talk about the wave function for a "pair."

Now the Schrodinger equation for the pair will be more or less like Eq. (21.3). There will be one difference in that the charge q will be twice the charge of an electron. Also, we don't know the inertia-or effective mass-for the pair in the crystal lattice, so we don't know what number to put in for m. Nor should we think that if we go to very high frequencies (or short wavelengths), this is exactly the right form, because the kinetic energy that corresponds to very rapidly varying wave functions may be so great as to break up the pairs. At finite temperatures there are always a few pairs which are broken up according to the usual Boltzmann theory. The probability that a pair is broken is proportional to $\exp(-E_{\text{pair}}/kT)$. The electrons that are not bound in pairs are called "normal" electrons and will move around in the crystal in the ordinary way. I will, however, consider only the situation at essentially zero tempearture -or, in any case, I will disregard the complications produced by those electrons which are not in pairs.

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[6] First discovered by Onnes in 1911; H. K. Onnes, Comm. Phys. Lab., Univ. Leyden, Nos. 119, 120, 122 (1911). You will find a nice up-to-date discussion of the subject in E. A. Lynton, *Superconductivity* John Wiley and Sons, Inc., New York, 1962.

[7] J. Bardeen, L. N. Cooper, and J. R. Schrieffer, *Phys. Rev.* **108,** 1175 (1957).

Since electron pairs are bosons. when there are a lot of them in a given state there is an especially large amplitude for other pairs to go to the same state. So nearly all of the pairs will be locked down at the lowest energy in *exactly the same* state-it won't be easy to get one of them into another state. There's more amplitude to go into the same state than into an unoccupied state by the famous factor $\sqrt{n}$, where n is the occupancy of the lowest state. So we would expect all the pairs to be moving in the same state.

What then will our theory look like? I'll call ψ the wave function of a pair in the lowest energy state. However, since $\psi\psi^*$ is going to be proportional to the charge density ρ, I can just as well write ψ as the square root of the charge density times some phase factor:

$$\psi(r) = \sqrt{\rho(r)}e^{i\theta(r)}, \tag{21.17}$$

where ρ and θ are real functions of r. (Any complex function can, of course, be written this way.) It's clear what we mean when we talk about the charge density, but what is the physical meaning of the phase θ of the wave function? Well, let's see what happens if we substitute $\psi(r)$ into Eq. (21.12) and express the current density in terms of these new variables ρ and θ. It's just a change of variables and I won't go through all the algebra, but it comes out

$$J = \frac{\hbar}{m}\left(\nabla\theta - \frac{q}{\hbar}A\right)\rho \tag{21.18}$$

Since both the current density and the charge density have a direct physical meaning for the superconducting electron gas, both ρ and θ are real things. The phase is just as observable as ρ; it is a piece of the current density J. The *absolute* phase is not observable, but if the gradient of the phase is known everywhere, the phase is known except for a constant. You can define the phase at one point, and then the phase everywhere is determined.

Incidentally, the equation for the current can be analyzed a little nicer, when you think that the current density J is *in fact* the charge density times the velocity of motion of the fluid of electrons, or ρv. Equation (21.18) is then equivalent to

$$mv = \hbar\,\nabla\theta - qA. \tag{21.19}$$

Notice that there are two pieces in the mv-momentum; one is a contribution from the vector potential, and the other, a contribution from the behavior of the wave function. In other words, the quantity $\hbar\nabla\theta$ is just what we have called the p-momentum.

21-6 The Meissner effect

Now we can describe some of the phenomena of superconductivity. First, there is no electrical resistance. There's no resistance because all the electrons are collectively in the same state. In the ordinary flow of current you knock one electron or the other out of the regular flow, gradually deteriorating the general momentum. But here to get one electron away from what all the others are doing is very hard because of the tendency of all Bose particles to go in the same state. A current once started, just keeps on going forever.

It's also easy to understand that if you have a piece of metal in the superconducting state and turn on a magnetic field which isn't too strong (we won't go into the details of how strong), the magnetic field can't penetrate the metal. If, as you build up the magnetic field, any of it were to build up inside the metal, there would be a rate of change of flux which would produce an electric field, and an electric field would immediately generate a current which, by Lenz's law, would oppose the flux. Since all the electrons will move together, an infinitesimal electric field will generate enough current to oppose completely any applied magnetic field. So if you turn the field on after you've cooled a metal to the superconducting state, it will be excluded.

Even more interesting is a related phenomenon discovered experimentally by **Meissner.**[8] If you have a piece of the metal at a high temperature (so that it is a normal conductor) and establish a magnetic field through it, and then you lower the temperature below the critical temperature (where the metal becomes a superconductor), *the field is expelled.* In other words, it starts up its own current-and in just the right amount to push the field out.

We can see the reason for that in the equations, and I'd like to explain how. Suppose that we take a piece of superconducting material which is in one lump. Then in a steady situation of any kind the divergence of the current must be zero because there's no place for it to go. It is convenient to choose to make the divergence of A equal to zero. (I should explain why choosing this convention doesn't mean any loss of generality, but I don't want to take the time.) Taking the divergence of Eq. (21.18), then gives that the Laplacian of θ is equal to zero. One moment. What about the variation of ρ? I forgot to mention an important point. There is a background of positive charge in this metal due to the atomic ions of the lattice. If the charge density ρ is uniform there is no net charge and no electric field. If there would be any accumulation of electrons in one region the charge wouldn't be neutralized and there would be a terrific repulsion pushing the electrons apart.† So in ordinary circumstances the charge density of the electrons in the superconductor is almost perfectly uniform-I can take ρ as a constant. Now the only way that $\nabla^2\theta$ can be zero everywhere inside the lump of metal is for θ to be a constant. And that means that there is no contribution to $\boldsymbol{J}$ from p-momentum. Equation (21.18) then says that the current is proportional to ρ times $\boldsymbol{A}$. So everywhere in a lump of superconducting material the current is necessarily proportional to the vector potential:

$$\boldsymbol{J} = -\rho\frac{q}{m}\boldsymbol{A}. \tag{21.20}$$

Since ρ and q have the same (negative) sign, and since ρ is a constant, I can set $\rho q/m = -$(some constant); then

$$\boldsymbol{J} = -(\text{some constant})\boldsymbol{A}. \tag{21.21}$$

This equation was originally proposed by London and **London**[9] to explain the experimental observations of superconductivity-long before the quantum mechanical origin of the effect was understood.

Now we can use Eq. (21.20) in the equations of electromagnetism to solve for the fields. The vector potential is related to the current density by

$$\nabla^2\boldsymbol{A} = -\frac{1}{\epsilon_0 c^2}\boldsymbol{J}. \tag{21.22}$$

If I use Eq. (21.21) for $\boldsymbol{J}$, I have

$$\nabla^2\boldsymbol{A} = \lambda^2\boldsymbol{A}, \tag{21.23}$$

where λ^2 is just a new constant;

$$\lambda^2 = \rho\frac{q}{\epsilon_0 mc^2}. \tag{21.24}$$

We can now try to solve this equation for $\boldsymbol{A}$ and see what happens in detail. For example, in one dimension Eq. (21.23) has exponential solutions of the form $e^{-\lambda x}$ and $e^{+\lambda x}$. These solutions mean that the vector potential must *decrease* exponentially as you go from the surface into the material. (It can't increase

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[8] W. Meissner and R. Ochsenfeld, *Naturwiss.* **21,** 787 (1933).

[9] H. London and F. London, *Proc. Roy. Soc.* (London) **A149,** 71 (1935); *Physica* **2,** 341 (1935).

† Actually if the electric field were too strong, pairs would be broken up and the "normal" electrons created would move in to help neutralize any excess of positive charge. Still, it takes energy to make these normal electrons, so the main point is that a nearly uniform density ρ is highly favored energetically.

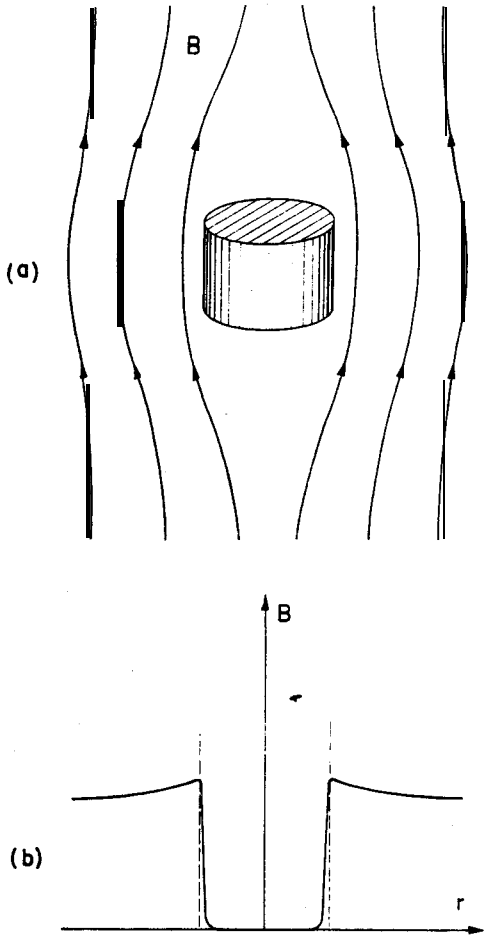


Fig. 21-3. (a) A superconducting cylinder is a magnetic field; (b) the magnetic field B as a function of r.

because there would be a blow up.) If the piece of metal is very large compared to $1/\lambda$. the field only penetrates to a thin layer at the surface-a layer about $1/\lambda$ in thickness. The entire remainder of the interior is free of field. as sketched in Fig. 21-3. This is the explanation of the Meissner effect.

How big is the distance λ? Well, remember that r_0, the "electromagnetic radius" of the electron (2.8×10^{-13} cm), is given by

$$mc^2 = \frac{q_e^2}{4\pi\epsilon_0 r_0}.$$

Also, remember that q in Eq. (21.24) is twice the charge on an electron, so

$$\frac{q}{\epsilon_0 mc^2} = \frac{8\pi r_0}{q_e},$$

Writing ρ as $q_e N$, where N is the number of electrons per cubic centimeter, we have

$$\lambda^2 = 8\pi N r_0. \tag{21.25}$$

For a metal such as lead there are about 3×10^{22} atoms per cm^3, so if each one contributed only one conduction electron, $1/\lambda$ would be about 2×10^{-5} cm. That gives you the order of magnitude.

21-7 Flux quantization

The London equation (21.21) was proposed to account for the observed facts of superconductivity including the Meissner effect. In recent times, however, there have been some even more dramatic predictions. One prediction made by London was so peculiar that nobody paid much attention to it until recently. I will now discuss it. This time instead of taking a single lump. suppose we take a *ring* whose thickness is large compared to $1/\lambda$, and try to see what would happen if we started with a magnetic field through the ring, then cooled it to the superconducting state, and afterward removed the original source of $\boldsymbol{B}$. The sequence of events is sketched in Fig. 21-4. In the normal state there will be a field in the body of the ring as sketched in part (a) of the figure. When the ring is made superconducting, the field is forced outside of the *material* (as we have just seen). There will then be some flux through the hole of the ring as sketched in part (b). If the external field is now removed, the lines of field going through the hole are "trapped" as shown in part (c). The flux Φ through the center can't decrease because $\partial\Phi/\partial t$ must be equal to the line integral of $\boldsymbol{E}$ around the ring, which is zero in a superconductor. As the external field is removed a super current starts flowing around the ring to keep the flux through the ring a constant. (It's the old eddy-current idea, only with zero resistance.) These currents will, however, all flow near the surface (down to a depth $1/\lambda$), as can be shown by the same kind of analysis that I made for the solid block. These currents can keep the magnetic field out of the body of the ring, and produce the permanently trapped magnetic field as well.

Now, however, there is an essential difference, and our equations predict a surprising effect. The argument I made above that θ must be a constant in a solid block does nor *apply for a ring,* as you can see from the following arguments.

Well inside the body of the ring the current density $\boldsymbol{J}$ is zero; so Eq. (21.18) gives

$$\hbar\nabla\theta = q\boldsymbol{A}. \tag{21.26}$$

Now consider what we get if we take the line integral of $\boldsymbol{A}$ around a curve Γ, which goes around the ring near the center of its cross-section so that it never gets near the surface. as drawn in Fig. 21-5. From Eq. (21.26)

$$\hbar\oint \nabla\theta \cdot d\boldsymbol{s} = q\oint \boldsymbol{A} \cdot d\boldsymbol{s}. \tag{21.27}$$

Now you know that the line integral of $\boldsymbol{A}$ around any loop is equal to the flux of $\boldsymbol{B}$ through the loop

$$\oint \boldsymbol{A} \cdot d\boldsymbol{s} = \Phi.$$

Equation (21.27) the becomes

$$\oint \nabla\theta \cdot d\boldsymbol{s} = \frac{q}{\hbar}\Phi. \tag{21.28}$$

The line integral of a gradient from one point to another (say from point 1 to point 2) is the difference of the values of the function at the two points. Namely,

$$\int_1^2 \nabla\theta \cdot d\boldsymbol{s} = \theta_2 - \theta_1.$$

If we let the two end points 1 and 2 come together to make a closed loop you might at first think that θ_2 would equal θ_1, so that the integral in Eq. (21.28) would be zero. That would be true for a closed loop in a simply-connected piece of superconductor, but it is not necessarily true for a ring-shaped piece. The only physical requirement we can make is that *there can be only one value of the wave function for each point.* Whatever θ does as you go around the ring, when you get back to the starting point the θ you get must give the same value for the wave function

$$\psi = \sqrt{\rho}e^{i\theta}.$$

This will happen if θ changes by $2\pi n$, where n is any integer. So if we make one complete turn around the ring the left-hand side of Eq. (21.27) must be $\hbar \cdot 2\pi n$. Using Eq. (21.28), I get that

$$2\pi n\hbar = q\Phi. \tag{21.29}$$

The trapped flux must always be an integer times $2\pi\hbar/q$*!* If you would think of the ring as a classical object with an ideally perfect (that is, infinite) conductivity, you would think that whatever flux was initially found through it would just stay there-any amount of flux at all could be trapped. But the quantum-mechanical theory of superconductivity says that the flux can be zero, or $2\pi\hbar/q$, or $4\pi\hbar/q$, or $6\pi\hbar/q$, and so on, but no value in between. It must be a multiple of a basic quantum mechanical unit.

London[10] predicted that the flux trapped by a superconducting ring would be quantized and said that the possible values of the flux would be given by Eq. (21.29) with q equal to the electronic charge. According to London the basic unit of flux should be $2\pi\hbar/q_e$, which is about 4×10^{-7} gauss = cm^2. To visualize such a flux, think of a tiny cylinder a tenth of a millimeter in diameter; the magnetic field inside it when it contains this amount of flux is about one percent of the earth's magnetic field. It should be possible to observe such a flux by a sensitive magnetic measurement.

In 1961 such a quantized flux was looked for and found by Deaver and Fairbank[11] at Stanford University and at about the same time by Doll and Nabauer [12] in Germany.

In the experiment of Deaver and Fairbank, a tiny cylinder of superconductor was made by electroplating a thin layer of tin on a one-centimeter length of No. 56 (1.3×10^{-3} cm diameter) copper wire. The tin becomes superconducting below 3.8°K, while the copper remains a normal metal. The wire was put in a small controlled magnetic field, and the temperature reduced until the tin became. superconducting. Then the exeternal source of field was removed. You would

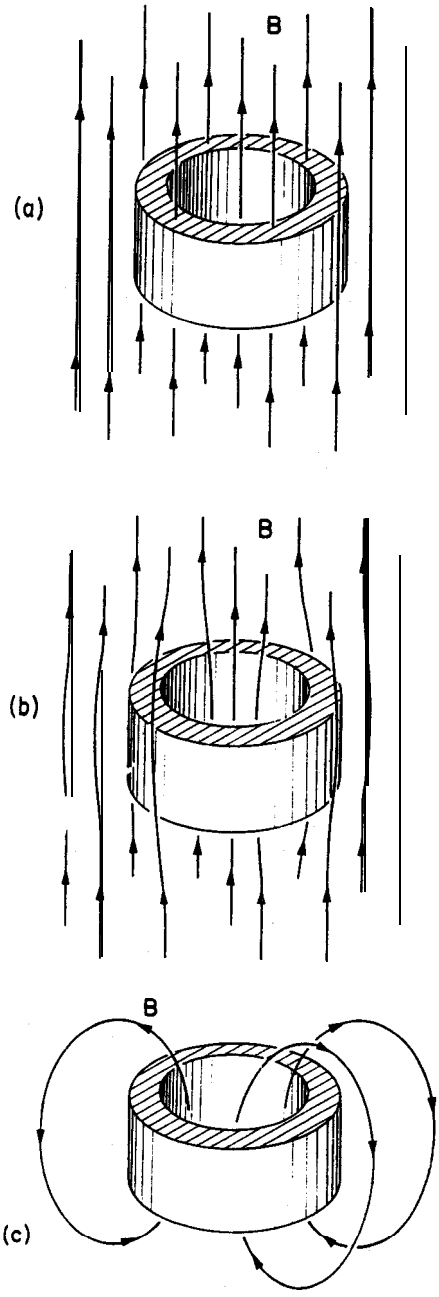


Fig. 21-4. A ring in a magnetic field: (a) in the normal state; (b) in the superconducting stote; (c) after the external field is removed.

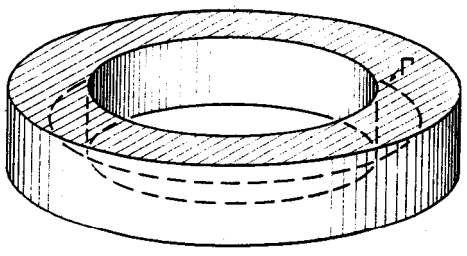


Fig. 21-5. The curve Γ inside a superconducting ring.

[10] F. London, *Superfluids;* John Wiley and Sons, Inc., New York, 1950, Vol. I, p. 152.
[11] B. S. Deaver, Jr., and W. M. Fairbank, *Phys Rev. Letters* **7,** 43 (1961).
[12] R. Doll and M. Nabauer, *Phys. Rev. Letters* **7,** 51 (1961).

expect this to generate a current by Lenz's law so that the flux inside would not change. The little cylinder should now have magnetic moment proportional to the flux inside. The magnetic moment was measured by jiggling the wire up and down (like the needle on a sewing machine, but at the rate of 100 cycles per second) inside a pair of little coils at the ends of the tin cylinder. The induced voltage in the coils was then a measure of the magnetic moment.

When the experiment was done by Deaver and Fairbank, they found that the flux was quantized, *but that the basic unit was only one-half as large as London had predicted.* Doll and Nabauer got the same result. At first this was quite mysterious,† but we now understand why it should be so. According to the Bardeen, Cooper, and Schrieffer theory of superconductivity, the q which appears in Eq. (21.29) is the charge of a *pair* of electrons and so is equal to $2q_e$. The basic flux unit is

$$\Phi_0 = \frac{\pi\hbar}{q_e} \approx 2 \times 10^{-7} \text{ gauss-cm} \tag{21.30}$$

or one-half the amount predicted by London. Everything now fits together, and the measurements show the existence of the predicted purely quantum-mechanical effect on a large scale.

21-8 The dynamics of superconductivity

The Meissner effect and the flux quantization are two confirmations of our general ideas. Just for the sake of completeness I would like to show you what the complete equations of a superconducting fluid would be from this point of view-it is rather interesting. Up to this point I have only put the expression for ψ into equations for charge density and current. If I put it into the complete Schrbdinger equation I get equations for ρ and θ. It should be interesting to see what develops, because here we have a "fluid" of electron pairs with a charge density ρ and a mysterious θ—we can try to see what kind of equations we get for such a "fluid"! So we substitute the wave function of Eq. (21.17) into the Schrodinger equation (21.3) and remember that ρ and θ are real functions of x, y, and z. If we separate real and imaginary parts we obtain then two equations. To write them in a shorter form I will-following Eq. (21.19)—write

$$\frac{\hbar}{m}\nabla\theta - \frac{q}{m}\boldsymbol{A} = \boldsymbol{v}. \tag{21.31}$$

One of the equations I get is then

$$\frac{\partial\rho}{\partial t} = \nabla \cdot \rho\boldsymbol{v}. \tag{21.32}$$

Since $\rho\boldsymbol{v}$ is first $\boldsymbol{J}$, this is just the continuity equation once more. The other equation I obtain tells how θ varies; it is

$$\hbar\frac{\partial\theta}{\partial t} = -\frac{m}{2}v^2 + q\phi - \frac{\hbar^2}{2m}\left\{\frac{1}{\sqrt{\rho}}\nabla^2(\sqrt{\rho})\right\}. \tag{21.33}$$

Those who are thoroughly familiar with hydrodynamics (of which I'm sure few of you are) will recognize this as the equation of motion for an electrically charged fluid if we identify $\hbar\theta$ as the "velocity potential"-except that the last term, which should be the energy of compression of the fluid, has a rather strange dependence on the density ρ. In any case, the equation says that rate of change of the quantity $\hbar\theta$ is given by a kinetic energy term, $\frac{1}{2}mv^2$, plus a potential energy term, $q\phi$, with an additional term, containing the factor $\hbar^2$, which we could call a "quantum mechanical energy." We have seen that inside a superconductor ρ is kept very

† It has once been suggested by Onsager that this might happen (see F. London, Ref. 10), although no one else ever understood why.

uniform by the electrostatic forces, so this term can almost certainly be neglected in every practical application provided we have only one superconducting region. If we have a boundary between two superconductors (or other circumstances in which the value of ρ may change rapidly) this term can become important.

For those who are not so familiar with the equations of hydrodynamics, I can rewrite Eq. (21.33) in a form that makes the physics more apparent by using Eq. (21.31) to express θ in terms of $\boldsymbol{v}$. Taking the gradient of the whole of Eq. (2 1.33) and expressing $\nabla\theta$ in terms of $\boldsymbol{A}$ and $\boldsymbol{v}$ by using (21.31), I get

$$\frac{\partial \boldsymbol{v}}{\partial t} = \frac{q}{m}\left(-\nabla\phi - \frac{\partial \boldsymbol{A}}{\partial t}\right) - \boldsymbol{v} \times (\nabla \times \boldsymbol{v}) - (\boldsymbol{v} \times \nabla)\boldsymbol{v} - \nabla\frac{\hbar^2}{2m}\left(\frac{1}{\sqrt{\rho}}\nabla^2\sqrt{\rho}\right). \tag{21.34}$$

What does this equation mean? First, remember that

$$-\nabla\phi - \frac{\partial \boldsymbol{A}}{\partial t} = \boldsymbol{E}. \tag{21.35}$$

Next, notice that if I take the curl of Eq. (21.19), I get

$$\nabla \times \boldsymbol{v} = -\frac{q}{m}\nabla \times \boldsymbol{A}, \tag{21.36}$$

since the curl of a gradient is always zero, But $\nabla \times \boldsymbol{A}$ is the magnetic field $\boldsymbol{B}$, so the first two terms can be written as

$$\frac{q}{m}(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}).$$

Finally, you should understand that $\partial\boldsymbol{v}/\partial t$ stands for the rate of change of the velocity of the fluid at a point. If you concentrate on a particular particle, its acceleration is the *total* derivative of $\boldsymbol{v}$ (or, as it is sometimes called in fluid dynamics, the "comoving acceleration"), which is related to $\partial\boldsymbol{v}/\partial t$ by[13]

$$\left.\frac{d\boldsymbol{v}}{dt}\right|_{\text{comoving}} = \frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v}\cdot\nabla)\boldsymbol{v}. \tag{21.37}$$

This extra term also appears as the third term on the right side of Eq. (21.25). Taking it to the left side, I can write Eq. (21.25) in the following way:

$$m\left.\frac{d\boldsymbol{v}}{dt}\right|_{\text{comoving}} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) - \nabla\frac{\hbar^2}{2}\left(\frac{1}{\sqrt{\rho}}\nabla^2\sqrt{\rho}\right). \tag{21.38}$$

We also have from Eq. (21.36) that

$$\nabla \times \boldsymbol{v} = -\frac{q}{m}\boldsymbol{B}. \tag{21.39}$$

These two equations are the equations of motion of the superconducting electron fluid. The first equation is just Newton's law for a charged fluid in an electromagnetic field. It says that the acceleration of each particle of the fluid whose charge is q comes from the ordinary Lorentz force $q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$ plus an additional force. which is the gradient of some mystical quantum mechanical potential-a force which is not very big except at the junction between two superconductors. The second equation says that the fluid is "ideal"--the curl of $\boldsymbol{v}$ has zero divergence (the divergence of $\boldsymbol{B}$ is always zero). That means that the velocity can be expressed in terms of velocity potential. Ordinarily one writes that $\nabla \times \boldsymbol{v} = 0$ for an ideal fluid, but for an *ideal charged fluid in a magnetic field,* this gets modified to Eq. (21.39).

So, Schrodinger's equation for the electron pairs in a superconductor gives us the equations of motion of an electrically charged ideal fluid. Superconductivity is the same as the problem of the hydrodynamics of a charged liquid. If you want

[13] See Volume II, Section 40-2.

to solve any problem about superconductors you take these equations for the fluid [or the equivalent pair, Eqs. (21.32) and (21.33)], and combine them with Maxwell's equations to get the fields. (The charges and currents you use to get the fields must, of course, include the ones from the superconductor as well as from the external sources.)

Incidentally, I believe that Eq. (21.38) is not quite correct, but ought to have an additional term involving the density. This new term does not depend on quantum mechanics, but comes from the ordinary energy associated with variations of density. Just as in an ordinary fluid there should be a potential energy density proportional to the square of the deviation of ρ from ρ_0, the undisturbed density (which is, here, also equal to the charge density of the crystal lattice). Since there will be forces proportional to the gradient of this energy, there should be another term in Eq. (21.38) of the form: (const) $\nabla(\rho - \rho_0)^2$. This term did not appear from the analysis because it comes from the interactions between particles, which I neglected in using an independent-particle approximation. It is, however, just the force I referred to when I made the qualitative statement that electrostatic forces would tend to keep ρ nearly constant inside a superconductor.

21-9 The Josephson junction

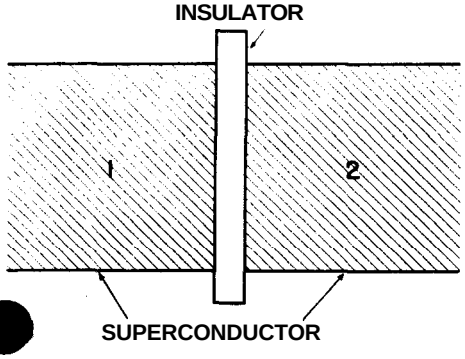


Fig. 21-6. Two superconductors separated by a thin insulator.

I would like to discuss next a very interesting situation that was noticed by Josephson [14] while analyzing what might happen at a junction between two superconductors. Suppose we have two superconductors which are connected by a thin layer of insulating material as in Fig. 21-6. Such an arrangement is now called a "Josephson junction." If the insulating layer is thick, the electrons can't get through; but if the layer is thin enough, there can be an appreciable quantum mechanical amplitude for electrons to jump across. This is just another example of the quantum-mechanical penetration of a barrier. Josephson analyzed this situation and discovered that a number of strange phenomenon should occur.

In order to analyze such a junction I'll call the amplitude to find an electron on one side, ψ_1, and the amplitude to find it on the other, ψ_2. In the superconducting state the wave function, ψ_1 is the common wave function of all the electrons on one side, and ψ_2 is the corresponding function on the other side. I could do this problem for different kinds of superconductors, but let us take a very simple situation in which the material is the same on both sides so that the junction is symmetrical and simple. Also, for a moment let there be no magnetic field. Then the two amplitudes should be related in the following way:

$$i\hbar \frac{\partial \psi_1}{\partial t} = U_1\psi_1 + K\psi_2,$$

$$i\hbar \frac{\partial \psi_2}{\partial t} = U_2\psi_2 + K\psi_1.$$

The constant K is a characteristic of the junction. If K were zero, these two equations would just describe the lowest energy state-with energy U-of each superconductor. But there is coupling between the two sides by the amplitude K that there may be leakage from one side to the other. (It is just the "flip-flop" amplitude of a two-state system.) If the two sides are identical, U_1 would equal U_2 and I could just subtract them off. But now suppose that we connect the two superconducting regions to the two terminals of a battery so that there is a potential difference V across the junction. Then $U_1 - U_2 = qV$. I can, for convenience, define the zero of energy to be halfway between, then the two equations are

$$i\hbar \frac{\partial \psi_1}{dt} = \frac{qV}{2}\psi_1 + K\psi_2,$$

$$i\hbar \frac{\partial \psi_2}{dt} = -\frac{qV}{2}\psi_2 + K\psi_1. \tag{21.40}$$

[14] B. D. Josephson, *Physics Letters* **1,** 251 (1962).

These are the standard equations for two quantum mechanical states coupled together. This time, let's analyze these equations in another way. Let's make the substitutions

$$\psi_1 = \sqrt{\rho_1}e^{i\theta_1}, \qquad \psi_2 = \sqrt{\rho_2}e^{i\theta_2}, \tag{21.41}$$

where θ_1 and θ_2 are the phases on the two sides of the junction and ρ_1 and ρ_2 are the density of electrons at those two points. Remember that in actual practice ρ_1 and ρ_2 are almost exactly the same and are equal to ρ_0, the normal density of electrons in the superconducting material. Now if you substitute these equations for ψ_1 and ψ_2 into (21.40) you get four equations by equating the real and imaginary parts in each case. Letting $(\theta_2 - \theta_1) = \delta$, for short, the result is

$$\dot{\rho}_1 = +\frac{2}{\hbar}K\sqrt{\rho_2\rho_1}\sin\delta, \tag{21.42}$$

$$\dot{\rho}_2 = -\frac{2}{\hbar}K\sqrt{\rho_2\rho_1}\sin\delta,$$

$$\dot{\theta}_1 = +\frac{K}{\hbar}\sqrt{\frac{\rho_2}{\rho_1}}\cos\delta - \frac{qV}{2\hbar}, \tag{21.43}$$

$$\dot{\theta}_2 = +\frac{K}{\hbar}\sqrt{\frac{\rho_1}{\rho_2}}\cos\delta + \frac{qV}{2\hbar}.$$

The first two equations say that $\dot{\rho}_1 = -\dot{\rho}_2$. "But," you say, "they must both be zero if ρ_1 and ρ_2 are both constant and equal to zero." Not quite. These equations are not the whole story. They say what $\dot{\rho}_1$ and $\dot{\rho}_2$ would be *if there were no extra electric forces* due to an unbalance between the electron fluid and the background of positive ions. They tell how the densities would *start* to change, and therefore describe the kind of current that would begin to flow. This current from side 1 to side 2 would be just $\dot{\rho}_1$(or $-\dot{\rho}_2$), or

$$J = \frac{2K}{\hbar}\sqrt{\rho_1\rho_2}\sin\delta. \tag{21.44}$$

Such a current would soon charge up side 2, *except* that we have forgotten that the two sides are connected by wires to the battery. The current that flows will not charge up region 2 (or discharge region 1) because currents will flow to keep the potential constant. These currents from the battery have not been included in our equations. When they are included, ρ_1 and ρ_2 do not in fact change, but the current across the junction is still given by Eq. (21.44).

Since ρ_1 and ρ_2 do remain constant and equal to ρ_0, let's set $2K\rho_0/\hbar = J_0$, and write

$$J = J_0\sin\delta. \tag{21.45}$$

J_0, like K, is then a number which is a characteristic of the particular junction.

The other pair of equations (21.43) tells us about θ_1 and θ_2. We are interested in the difference $\delta = \theta_2 - \theta_1$ to use Eq. (21.45); what we get is

$$\dot{\delta} = \dot{\theta}_2 - \dot{\theta}_1 = \frac{qV}{\hbar}. \tag{21.46}$$

That means that we can write

$$\delta(t) = \delta_0 + \frac{q}{\hbar}\int V(t)\,dt, \tag{21.47}$$

where δ_0 is the value of δ at $t = 0$. Remember also that q is the charge of a pair, namely, $q = 2q_e$. In Eqs. (21.45) and (21.47) we have an important result, the general theory of the Josephson junction.

Now what are the consequences? First, put on a dc voltage. If you put on a dc voltage, V_0, the argument of the sine becomes $(\delta_0 + (q/\hbar)V_0 t)$. Since $\hbar$ is a small number (compared to ordinary voltage and times), the sine oscillates rather rapidly and the net current is nothing. (In practice, since the temperature is not zero, you would get a small current due to the conduction by "normal" electrons.) On the other hand if you have *zero* voltage across the junction, you can get a current! With no voltage the current can be any amount between $+J_0$ and $-J_0$ (depending on the value of δ_0). But try to put a voltage across it and the current goes to zero. This strange behavior has recently been observed experimentally.[15]

There is another way of getting a current-by applying a voltage at a very high frequency in addition to a dc voltage. Let

$$V = V_0 + v \cos \omega t,$$

where $v \ll V$. Then $\delta(t)$ is

$$\delta_0 + \frac{q}{\hbar} V_0 t + \frac{q}{\hbar} \frac{v}{\omega} \sin \omega t.$$

Now for Δx small,

$$\sin (x + \Delta x) \approx \sin x + \Delta x \cos x.$$

Using this approximation for sin δ, I get

$$J = J_0 \left[\sin \left(\delta_0 + \frac{q}{\hbar} V_0 t \right) + \frac{q}{\hbar} \frac{v}{\omega} \sin \omega t \cos \left(\delta_0 + \frac{q}{\hbar} V_0 t \right) \right].$$

The first term is zero on the average, but the second term is not if

$$\omega = \frac{q}{\hbar} V_0.$$

There should be a current if the ac voltage has just this frequency. Shapiro[16] claims to have observed such a resonance effect.

If you look up papers on the subject you will find that they often write the formula for the current as

$$J = J_0 \sin \left(\delta_0 + \frac{2q_e}{\hbar} \int A \cdot ds \right), \tag{21.48}$$

where the integral is to be taken across the junction. The reason for this is that when there's a vector potential across the junction the flip-flop amplitude is modified in phase in the way that we explained earlier. If you chase that extra phase through, it comes out as given above.

Finally, I would like to describe a very dramatic and interesting experiment which has recently been made on the interference of the currents from each of two junctions. In quantum mechanics we're used to the interference between amplitudes from two different slits. Now we're going to do the interference between two junctions caused by the difference in the phase of the arrival of the currents through two different paths. In Fig. 21-7, I show two different junctions, "a" and "b", connected in parallel. The ends, P and Q, are connected to our electrical intruments which measure any current flow. The external current, J_{total}, will be the sum of the currents through the two junctions. Let J_a and J_b be the currents through the two junctions, and let their phases be δ_a and δ_b. Now the phase difference of the wave functions between P and Q must be the same whether you go on one route or the other. Along the route through junction "a", the phase difference between P and Q is δ_a plus the line integral of the vector potential along the upper route:

$$\Delta\text{Phase}_{P \to Q} = \delta_a + \frac{2q_e}{\hbar} \int_{\text{upper}} A \cdot ds. \tag{21.49}$$

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[15] P. W. Anderson and J. M. Rowell, *Phys. Rev. Letters* **10,** 230 (1963).
[16] S. Shapiro, *Phys. Rev. Letters* **11,** 80 (1963).

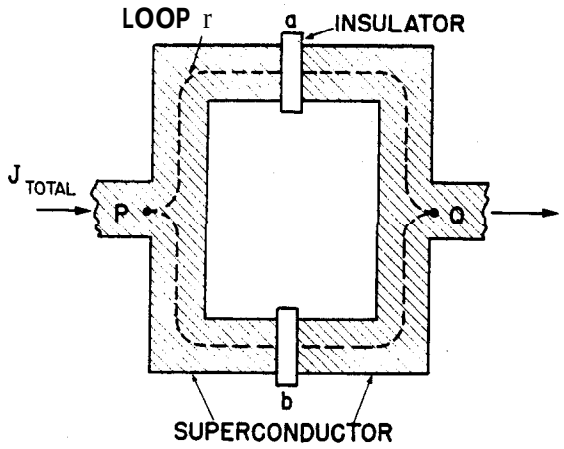


Fig. 21-7. Two Josephson junctions in parallel.

Why? Because the phase θ is related to $\boldsymbol{A}$ by Eq. (21.26). If you integrate that equation along some path, the left-hand side gives the phase change, which is then just proportional to the line integral of $\boldsymbol{A}$, as we have written here. The phase change along the lower route can be written similarly

$$\Delta\text{Phase}_{P\to Q} = \delta_b + \frac{2q_e}{\hbar}\int_{\text{lower}} \boldsymbol{A}\cdot d\boldsymbol{s}. \tag{21.50}$$

These two must be equal; and if I subtract them I get that the difference of the deltas must be the line integral of $\boldsymbol{A}$ around the circuit:

$$\delta_b - \delta_a = \frac{2q_e}{\hbar}\oint_\Gamma \boldsymbol{A}\cdot d\boldsymbol{s}.$$

Here the integral is around the closed loop Γ of Fig. 21-7 which circles through both junctions. The integral over $\boldsymbol{A}$ is the magnetic flux Φ through the loop. So the two δ's are going to differ by $2q_e/\hbar$ times the magnetic flux Φ which passes between the two branches of the circuit:

$$\delta_b - \delta_a = \frac{2q_e}{\hbar}\Phi. \tag{21.51}$$

I can control this phase difference by changing the magnetic field on the circuit, so I can adjust the differences in phases and see whether or not the total current that flows through the two junctions shows any interference of the two parts. The total current will be the sum of J_a and J_b. For convenience, I will write

$$\delta_a = \delta_0 + \frac{q_e}{\hbar}\Phi, \qquad \delta_b = \delta_0 - \frac{q_e}{\hbar}\Phi.$$

Then,

$$J_{\text{total}} = J_0\left(\sin\left(\delta_0 + \frac{q_e}{\hbar}\Phi\right) + \sin\left(\delta_0 - \frac{q_e}{\hbar}\Phi\right)\right)$$

$$= 2J_0 \sin\delta_0 \cos\frac{q_e\Phi}{\hbar}. \tag{21.52}$$

Now we don't know anything about δ_0, and nature can adjust that anyway she wants depending on the circumstances. In particular, it will depend on the external voltage we apply to the junction. No matter what we do, however, $\sin\delta_0$ can never get bigger than 1. So the *maximum* current for any given Φ is given by

$$J_{\max} = J_0\left|\cos\frac{q_e\Phi}{\hbar}\right|.$$

This maximum current will vary with Φ and will itself have maxima whenever

$$\Phi = n\frac{\pi\hbar}{q_e},$$

with n some integer. That is to say that the current takes on its maximum values where the flux linkage has just those quantized values we found in Eq. (21.30)!

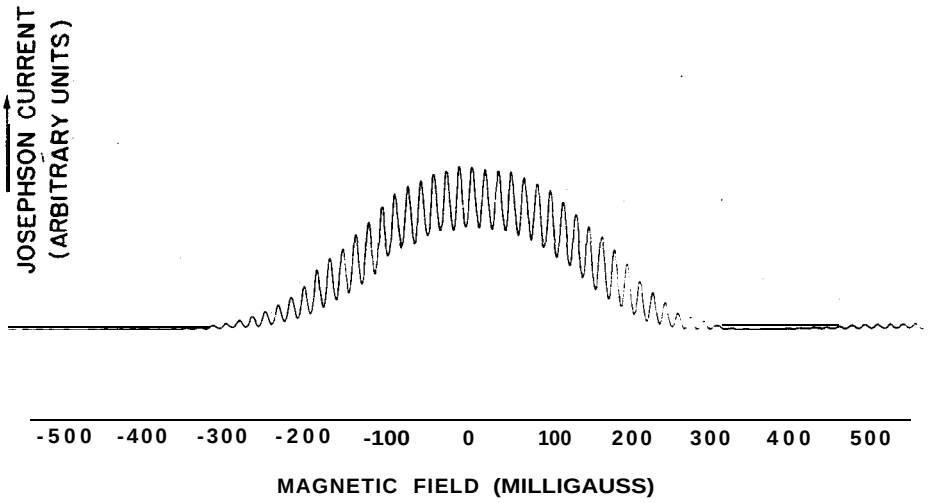


Fig. 21-8. A recording of the current through a pair of Josephson junctions as a function of the magnetic field in the region between the two junctions (see Fig. 21-7). [This recording was provided by R. C. Jaklevic, J. Lambe, A. H. Silver, and J. E. Mercereau of the Scientific Laboratory, Ford Motor Company.]

The Josephson current through a double junction was recently measured [17] as a function of the magnetic field in the area between the junctions. The results are shown in Fig. 21-8. There is a general background of current from various effects we have neglected, but the rapid oscillations of the current with changes in the magnetic field are due to the interference term cos $q_e\Phi/\hbar$ of Eq. (21.52).

One of the intriguing questions about quantum mechanics is the question of whether the vector potential exists in a place where there's no field.[18] This experiment I have just described has also been done with a tiny solenoid between the two junctions so that the only significant magnetic $\boldsymbol{B}$ field is inside the solenoid and a negligible amount is on the superconducting wires themselves. Yet it is reported that the amount of current depends oscillatorily on the flux of magnetic field inside that solenoid even though that field never touches the wires-another demonstration of the "physical reality" of the vector potential.[19]

I don't know what will come next. But look what can be done. First, notice that the interference between two junctions can be used to make a sensitive magnetometer. If a pair of junctions is made with an enclosed area of, say, 1 mm^2. the maxima in the curve of Fig. 21-8 would be separated by 2×10^{-6} gauss. It is certainly possible to tell when you are 1/10 of the way between two peaks; so it should be possible to use such a junction to measure magnetic fields as small as 2×10^{-7} gauss-or to measure larger fields to such a precision. One should be able to go even farther. Suppose for example we put a set of 10 or 20 junctions close together and equally spaced. Then we can have the interference between 10 or 20 slits and as we change the magnetic field we will get very sharp maxima and minima. Instead of a 2-slit interference we can have a 20- or perhaps even a 100-slit interferometer for measuring the magnetic field. Perhaps we can predict that the measurement of magnetic fields will-by using the effects of quantum-mechanical interference-eventually become almost as precise as the measurement of wavelength of light.

These then are some illustrations of things that are happening in modern times-the transistor, the laser, and now these junctions, whose ultimate practical applications are still not known. The quantum mechanics which was discovered in 1926 has had nearly 40 years of development, and rather suddenly it has begun to be exploited in many practical and real ways. We are really getting control of nature on a very delicate and beautiful level.

I am sorry to say, gentlemen, that to participate in this adventure it is absolutely imperative that you learn quantum mechanics as soon as possible. It was our hope that in this course we would find a way to make comprehensible to you at the earliest possible moment the mysteries of this part of physics.

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[17] Jaklevic, Lambe, Silver, and Mercereau, *Phys. Rev. Letters* **12,** 159 (1964).
[18] Jaklevic, Lambe, Silva, and Mercereau, *Phys. Rev. Letters* **12,** 274 (1964).
[19] See Volume II, Chapter 15, Section 15-5.